



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

First Semester M.Tech. (CST) (Credit System) Degree Examinations

December 2024 – January 2025

22CST131 – REPAIR AND RESTORATION OF STRUCTURES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit I and Unit II and One full question from Unit III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the application of ultrasonic Pulse Velocity Test with a neat diagram.	10	L*2	1	1
b)	Briefly explain about the effect of chemicals on concrete structures.	10	L2	2	1
2. a)	Illustrate the holistic model of deterioration of concrete structures with the help of a Venn Diagram.	10	L2	1	1
b)	Explain the corrosion mechanism with the help of a neat sketch.	10	L2	2	1
3. a)	Explain the factors affecting the durability of concrete structures with the help of a flowchart.	10	L2	1	1
b)	Explain in brief design errors in concrete structures.	10	L2	2	1
Unit – II					
4. a)	Explain in brief the various factors that affect the concrete structure which necessitates the maintenance.	10	L2	3	1
b)	Explain in detail with a neat flow chart types and causes of cracks in hardened concrete.	10	L2	4	1
5. a)	Explain with the help of flow chart the assessment procedure adapted for concrete structures.	10	L2	3	1
b)	Explain the application of rust eliminators in repair of damaged buildings.	10	L2	4	1
6. a)	Provide the classification of maintenance work and explain about each of them briefly.	10	L2	3	1
b)	Explain in detail about Guniting and shotcrete.	10	L2	4	1
Unit – III					
7. a)	Explain with the help of neat sketches the strengthening of concrete columns by fiber wraps.	20	L3	5	2
8. a)	Explain with neat sketches the method of strengthening of beam by				
i.	Adding new member				
ii.	Section enlargement	20	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
